Vanagarouthu Sree Chaitran

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Education

National Institute of Technology, Delhi **Aug. 2025 – May. 2027**
M.Tech in Mathematics and Computing *GPA: 7.25/10*

Vellore Institute of Technology, Vellore **Sep. 2020 – May. 2024**
B.Tech in CSE with IoT *GPA: 7.86/10*

Experience

Artificial Intelligence Internship **May 2023 – July 2023**
Smart Bridge

- Movie Recommendation Based on Emotion Using Web Scraping with IBM.
- Collected and processed large volumes of movie data from multiple online sources using Python-based web scraping techniques to support sentiment and emotion analysis.
- Developed an emotion-driven movie recommendation model integrating NLP and sentiment analysis, enabling personalized movie suggestions aligned with the user's emotional state.
- Designed an interactive Flask web interface to deliver real-time recommendations, improving user engagement and enhancing the overall recommendation experience.

Projects

Seasonal Arctic Sea-Ice Forecasting Using a Swin-UNet Deep Learning Model | *GitHub*

- Built an end-to-end Swin-UNet transformer pipeline forecasting Arctic sea-ice concentration up to 6 months ahead, from NSIDC satellite and PIOMAS thickness data processed into an 18-channel spatiotemporal input on a 96×96 polar grid.
- Trained a 60-model ensemble (20-fold leave-one-year-out × 3 seeds) and showed that sea-ice-thickness inputs significantly improve forecast skill (p=0.031), using bootstrap resampling and FDR-corrected significance tests.
- Benchmarked against CNN U-Net baselines with matched (15M) and larger (140M) parameter counts, separating the gains from architecture from the gains from model size.

Agentic Restaurant Order Management System using LangGraph and LLMs | *GitHub*

- Built a stateful multi-agent workflow in **LangGraph** with 9 nodes and conditional routing. It takes an order, checks stock, then handles cooking, serving, retries and failures through one typed state.
- Used an **LLM (Groq, GPT-OSS-120B)** to pull the dish and quantity out of free-text orders as structured JSON. A guardrail rejects requests that aren't food orders, and the LLM writes the replies to the customer.
- Added inventory checks (fully / partly / not available) with separate retry limits for ordering (3), cooking (2) and serving (2). Cooking and serving fail at random 40% of the time. A scenario-based test harness checks the exact path through the graph; all 3 end-to-end test cases passed.

Vector Database and Local RAG System Using C++ and Ollama |

- Implemented a C++ vector database supporting HNSW, KD-Tree, and Brute Force similarity search with cosine, Euclidean, and Manhattan distance metrics.
- Integrated Ollama-based embeddings and local LLM generation to build a document-based RAG pipeline for semantic search and question answering.
- Exposed REST APIs for vector CRUD operations, benchmarking, document insertion, retrieval, and AI-powered responses, with a web UI for visualization and interaction.

Journal Paper

Lane Prediction Optimization and Secure Traffic Management Using VANET **Harvest Journal, 2023**

Technical Skills

Languages: C++, Python, MATLAB, JavaScript, PHP, SQL, HTML, CSS
AI/ML: Machine Learning, Deep Learning, NLP, Vector Embeddings, RAG, Semantic Search
Frameworks & Libraries: Flask, React, Matplotlib, AJAX, Ollama, REST APIs
Systems & Tools: Git, VS Code, Google Cloud Platform, Vector Databases, HNSW, KD-Tree

Achievements / Extracurricular

- Placement Coordinator, NIT Delhi **Jan 2026- May 2027**
- Core Committee Member – Nature Lovers Club, VIT Vellore **Jan 2023- Jan 2024**
- Achieved a rank of 6160 out of a competitive pool of 1,43,265 candidates in the TS EAMCET. **May 2020**